Demographic Analysis of Behavioral Health Service Utilization Among Children and Youth

Objectives:

1. **Identify Underrepresented Demographics**
   Explore why certain ethnic groups may be underrepresented in behavioral health services, examining barriers and comparing utilization data to regional population statistics.
2. **Investigate Gender Disparities**
   Analyze gender-based differences in service utilization, focusing on specific services accessed by male and female users and their corresponding outcomes.
3. **Examine Age Distribution**
   Identify patterns in service utilization across age groups, assessing which groups are over- or underrepresented and their service preferences.
4. **Analyze Regional Variations**
   Compare the demographic profiles of service users in urban and rural regions, investigating the role of policies or resources in shaping regional differences.
5. **Track Temporal Changes in Demographics**
   Assess shifts in the demographic composition of service users over time, linking trends to potential policy or societal changes.
6. **Perform Cross-Tabulation Analyses**
   Explore the intersection of demographic variables, such as age and ethnicity, to uncover nuanced insights into under- or overrepresented subgroups.
7. **Assess Socioeconomic Factors**
   Investigate the influence of socioeconomic indicators on service utilization and identify barriers faced by specific demographic groups.

Introduction:

1. File Loading and Mapping

- **Lists Files in the Working Directory:**
  - The `os.listdir()` function lists all files in the current working directory.
  - This is useful for identifying which datasets are available for processing.
- **Maps File Names to Descriptive Dataset Names:**
  - A dictionary (`file_mapping`) is used to map specific file names to more meaningful dataset names.
  - For example, the file `children-and-youth-demographics-by-age-group.csv` is mapped to `demographics_age_group`.

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2. Loading the Datasets

- **Loads Files into DataFrames:**
  - Each CSV file in the directory that matches a key in `file_mapping` is loaded into a pandas DataFrame.
  - JSON files are loaded using Python's `json` library.
  - The loaded datasets are stored in a dictionary (`datasets`) with descriptive names as keys.
- **Confirms Successful Loading:**
  - Prints a success message for each file loaded and previews the first few rows using `df.head()` to verify the data.

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3. Data Cleaning

For each dataset loaded into a DataFrame:

- **Handles Missing Values:**
  - Categorical columns (`object` type) have missing values filled with the most frequent value (mode).
  - Numerical columns have missing values filled with the median.
- **Removes Duplicates:**
  - Removes any duplicate rows in the dataset to ensure data integrity.
- **Standardizes Column Names:**
  - Column names are converted to lowercase and spaces are replaced with underscores for consistency and easier referencing.
- **Optional Data Type Conversion:**
  - The code includes a placeholder to convert date columns to datetime format (commented out for now).

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4. Saving Cleaned Datasets

- Each cleaned DataFrame is saved as a new CSV file with the format `<dataset_name>_cleaned.csv`. This ensures the original files remain unaltered.

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5. Displaying Cleaned Dataset Information

- After cleaning, the code optionally displays the structure (`df.info()`) of each cleaned DataFrame, summarizing the number of rows, columns, and data types.

Purpose of the Code in Context (Snippet #1) :

This code is designed to prepare raw data for analysis. The cleaned and standardized datasets can be used for the following analyses outlined in the assignment scenarios:

1. Investigating disparities in behavioral health service utilization.
2. Examining demographic trends and patterns across various dimensions like age, race, gender, and region.
3. Identifying correlations between demographic variables and behavioral health outcomes.

The robust cleaning steps ensure that the datasets are free of duplicates, missing values, and inconsistent formatting, allowing for accurate and meaningful analysis.

Code Explanation in Reference to Scenario 1: Identifying Underrepresented Demographics (Snippet #2)

Scenario Context:

In Scenario 1, we are tasked with analyzing disparities in behavioral health service utilization among different ethnic groups. The goal is to determine if specific groups are underrepresented in service usage compared to their population proportions, explore potential barriers, and propose actionable recommendations to address these disparities.

Code Breakdown

1. Load and Prepare the Data

- **Loading Data:**
  - The code attempts to load two datasets:
    1. `utilization_race_group.csv`: Contains the number of service users categorized by race.
    2. `demographics_race_group.csv`: Contains the population data categorized by race.
  - If the files are missing, sample data is used as a fallback to ensure functionality.
- **Column Standardization:**
  - Ensures column names are consistent (lowercase and underscores instead of spaces) for easier manipulation and merging.

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2. Calculate Proportions

- **Service Utilization Proportion:**
  - The proportion of each racial group's service users is calculated by dividing the number of service users by the total number of service users.
- **Population Proportion:**
  - Similarly, the proportion of each racial group within the total population is calculated.

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3. Merge Datasets

- The two datasets are merged on the `race` column, creating a single dataset (`merged_data`) with both utilization and population proportions for each racial group.

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4. Calculate Utilization-to-Population Ratios

- **Normalization of Utilization Rates:**
  - A ratio is computed to compare service utilization to the population proportion for each racial group: Utilization-to-Population Ratio=Utilization ProportionPopulation Proportion\text{Utilization-to-Population Ratio} = \frac{\text{Utilization Proportion}}{\text{Population Proportion}}Utilization-to-Population Ratio=Population ProportionUtilization Proportion
  - **Interpretation:**
    - A ratio of **1** indicates parity (the racial group uses services in proportion to their population).
    - A ratio **below 1** indicates **underrepresentation** (the group uses fewer services than expected based on their population size).
    - A ratio **above 1** indicates **overrepresentation**.

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5. Identify Underrepresented Groups

- The code flags racial groups with a utilization-to-population ratio below 1 as underrepresented.
- This step is critical for understanding disparities in service access.

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6. Visualization

Two key visualizations are created to illustrate the findings:

1. **Population vs. Utilization Proportions (Bar Plot):**
    - This plot compares the population proportion of each racial group to their proportion of service users.
    - It highlights disparities between the two metrics.
2. **Utilization-to-Population Ratios (Bar Plot):**
    - This plot visualizes the ratios for each racial group.
    - Bars are color-coded:
        - **Red**: Underrepresented groups (ratio < 1).
        - **Green**: Adequately represented or overrepresented groups (ratio ≥ 1).

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7. Hypotheses and Recommendations

- **Hypotheses:**
    - For underrepresented groups, the code suggests potential barriers such as:
        1. **Language barriers**: Difficulty communicating with service providers.
        2. **Cultural stigmas**: Negative perceptions of mental health services.
        3. **Service availability**: Lack of nearby facilities in underserved areas.
        4. **Socioeconomic factors**: Financial constraints or lack of insurance.

- **Recommendations:**
    - The code proposes actions to address disparities, such as:
        1. **Awareness Campaigns**: Educate underrepresented communities about available services.
        2. **Multilingual Services**: Hire staff fluent in multiple languages to overcome communication barriers.
        3. **Culturally Sensitive Programs**: Reduce stigma around mental health services through culturally tailored outreach.
        4. **Improve Accessibility**: Establish services in underserved regions to increase availability.

Relevance to Scenario 1

The code directly addresses the objectives of Scenario 1 by:

1. **Quantifying Disparities:**
    - Identifies racial groups that are underrepresented in behavioral health service utilization relative to their population size.
2. **Providing Insights:**
    - Visualizations and calculations help clearly communicate which groups are underrepresented and the magnitude of the disparity.

3. **Laying the Foundation for Action:**
    - The hypotheses and recommendations offer practical steps to address underrepresentation and improve equity in service access.

Key Insights for Scenario 1

For instance, using the sample data:

- The **Hispanic** group might show underrepresentation with a ratio significantly below 1.
- The analysis highlights the need for targeted interventions to address barriers specific to this group.

The overall approach ensures that disparities are not only identified but also contextualized with actionable recommendations, aligning perfectly with the goals of Scenario 1.

Code Explanation in Reference to Scenario 2: Gender Disparities in Behavioral Health Services (Snippet #3)

Scenario Context:

Scenario 2 focuses on analyzing the gender disparities in the utilization of behavioral health services. The goal is to understand:

1. The total service usage by gender.
2. Differences in service type utilization (e.g., therapy, counseling) between genders.
3. Service outcomes (e.g., positive or negative) by gender.
4. Gender-specific percentage distributions for deeper insights.

This analysis identifies patterns or inequalities in service access, types, and outcomes, which can inform strategies to promote gender equity in behavioral health services.

Code Breakdown

1. Dataset Setup

- **Input Data:**
    - The dataset contains columns:
        - **`gender`**: Gender of the individual (`Male` or `Female`).
        - **`service_type`**: Type of service used (e.g., `Therapy`, `Counseling`, `Medication`).
        - **`outcome`**: The outcome of the service (`Positive` or `Negative`).
        - **`count`**: Number of cases for each combination of gender, service type, and outcome.

- **Purpose:**
  - Analyze how behavioral health services differ in terms of utilization and outcomes by gender.

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2. Total Service Utilization by Gender

- **Analysis:**
  - The total number of services used by males and females is computed by summing the `count` column for each gender.
- **Visualization:**
  - A **bar chart** displays the total service utilization by gender:
    - **Blue** for males and **pink** for females.
    - This visually highlights any disparities in overall service usage.

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3. Service Type Utilization by Gender

- **Analysis:**
  - The dataset is grouped by both `gender` and `service_type`.
  - The total count of services for each gender-service combination is calculated and presented in a pivot table.
- **Visualization:**
  - A **stacked bar chart** or grouped bar chart shows the utilization of each service type (`Therapy`, `Counseling`, `Medication`) by gender:
    - This allows comparisons of how men and women utilize different types of services.

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4. Service Outcomes by Gender

- **Analysis:**
  - The dataset is grouped by `gender` and `outcome`.
  - The total count of positive and negative outcomes for each gender is computed and presented in a pivot table.
- **Visualization:**
  - A **stacked bar chart** illustrates the distribution of service outcomes by gender:
    - Each bar represents a gender, split into positive and negative outcomes.
    - The chart highlights whether one gender experiences disproportionately more positive or negative outcomes.

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5. Deeper Dive: Percentage Analysis

- **Purpose:**
  - This deeper analysis helps in understanding proportional differences between genders rather than just absolute numbers.
- **Output:**
  - The percentages are printed in the console for further interpretation.

How the Code Addresses Scenario 2

This code answers Scenario 2's objectives comprehensively:

1. **Total Service Utilization (Gender-Based Disparities):**
   - Highlights whether males or females use behavioral health services more frequently.
   - Example Insight: Males might use more services overall, but further analysis reveals disparities in service types or outcomes.
2. **Service Type Utilization (Gender Preferences):**
   - Examines differences in service type preferences (e.g., men may prefer `Medication`, while women might use more `Therapy`).
   - Example Insight: Men might underutilize counseling services compared to women, signaling a need for targeted outreach.
3. **Service Outcomes (Quality of Services):**
   - Identifies whether one gender experiences fewer positive outcomes, indicating potential issues with service quality, accessibility, or fit.
   - Example Insight: If females have more negative outcomes in therapy, this may suggest a need for gender-specific interventions.
4. **Percentage Analysis (Proportional View):**
   - Helps normalize the data for differences in total counts, revealing whether disparities exist even when controlling for total usage.
   - Example Insight: While men might have a higher absolute count for medication usage, women might have a higher proportional usage of this service type.

Example Insights from the Code (Based on Sample Data)

1. **Total Service Utilization:**
   - Males use 270 services compared to 170 for females.
   - Insight: Males may access behavioral health services more frequently.
2. **Service Type Utilization:**
   - Males primarily use therapy (120 cases) and counseling (100 cases), with fewer medication cases (50).

    - Females use therapy (80 cases), counseling (60 cases), and fewer medication cases (30).
    - Insight: Counseling is less utilized by females, indicating a possible need for increased awareness or availability.
3. **Service Outcomes:**
    - Males have 270 total outcomes (220 positive, 50 negative).
    - Females have 170 total outcomes (120 positive, 50 negative).
    - Insight: While the overall percentage of positive outcomes is higher for males, the percentage of negative outcomes is the same for both genders (approximately 18.5%).
4. **Percentage Analysis:**
    - Males:
        - Therapy: **44%**, Counseling: **37%**, Medication: **19%**.
    - Females:
        - Therapy: **47%**, Counseling: **35%**, Medication: **18%**.
    - Insight: Both genders have a similar proportional distribution, but males slightly favor counseling over therapy compared to females.

Conclusion and Relevance to Scenario 2

This code provides detailed insights into gender disparities in behavioral health services. It identifies patterns in service access, preferences, and outcomes. The analysis can inform targeted interventions, such as:

- Promoting services underutilized by a specific gender (e.g., counseling for females).
- Improving service outcomes for genders with higher negative results.
- Developing gender-sensitive programs to address specific needs.

These insights align with Scenario 2's goal of understanding and addressing gender-based disparities in behavioral health services.

Code Explanation in Reference to Scenario 3: Age Distribution of Service Users (Snippet #4)

Scenario Context:

Scenario 3 aims to analyze the age distribution of children and youth who utilize mental health services. The objective is to categorize users into different age groups (e.g., 0-5, 6-12, 13-18 years) to understand which age ranges have higher or lower service utilization. This information can be used to design age-specific interventions, allocate resources effectively, and identify underserved age groups.

Code Walkthrough

1. Dataset Loading and Cleaning

- **Dataset Files:**
  - Multiple CSV and JSON files related to children and youth mental health services are assumed to be uploaded.
  - The code lists the files in the working directory and identifies datasets using a predefined mapping (`file_mapping`) to assign descriptive names to the files.
- **Loading Data:**
  - Each file is loaded into a pandas DataFrame (for CSV files) or a dictionary (for JSON files).
  - Successfully loaded datasets are stored in a `datasets` dictionary for easy access.
- **Data Cleaning:**
  - **Missing Values:**
    - Categorical columns: Missing values are filled with the most frequent value (mode).
    - Numerical columns: Missing values are filled with the median value.
  - **Duplicate Removal:**
    - Duplicate rows are removed to ensure data accuracy.
  - **Column Renaming:**
    - Column names are standardized to lowercase and spaces are replaced with underscores for consistency.
  - **Saving Cleaned Data:**
    - Each cleaned dataset is saved as a new CSV file for future use.
- **Purpose:**
  - Ensures the data is ready for analysis by handling inconsistencies, missing values, and formatting issues.

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2. Age Distribution Analysis

- The analysis focuses on the **`utilization_age_group`** dataset, which contains age-related information about service users.

Step 1: Verify the Dataset

- **Column Inspection:**
  - The code prints the column names and previews the first few rows to ensure the dataset structure aligns with the analysis requirements.
  - The dataset is checked for the presence of an age-related column (e.g., `age` or `age_group`).

Step 2: Define Age Groups

- **Age Bins:**
  - Age ranges are defined using bins: `[0, 5, 12, 18]` to create three categories:
    - `0-5 years`
    - `6-12 years`
    - `13-18 years`
  - Labels (`age_labels`) are assigned to these bins.
- **Purpose:**
  - Helps classify users into meaningful age categories for comparison.

Step 3: Categorize Users into Age Groups

- The **`pd.cut()`** function is used to categorize individuals into the defined age groups based on their age values.
- A new column, **`age_group`**, is added to the dataset, indicating each user's corresponding age category.

Step 4: Count Users in Each Age Group

- **Value Counts:**
  - The number of users in each age group is calculated using **`value_counts()`** and sorted by age group order.
- **Purpose:**
  - Identifies which age groups have higher or lower utilization of mental health services.

Step 5: Visualize Age Distribution

- **Bar Plot:**
  - The age group distribution is visualized using a Seaborn bar plot:
    - **X-Axis:** Age groups (`0-5`, `6-12`, `13-18`).
    - **Y-Axis:** Number of users in each age group.
    - **Color Palette:** A `viridis` palette is applied for visual appeal.
  - **Title and Labels:**
    - The plot is titled "Age Distribution of Service Users."
    - Axes are labeled for clarity.
- **Purpose:**
  - Provides a clear visualization of age group disparities in service utilization.

Relevance to Scenario 3

This code addresses Scenario 3's goal of analyzing the **age distribution of service users**. Here's how the analysis applies:

1. **Categorizing Users by Age:**
   - The code creates meaningful age groups (`0-5`, `6-12`, `13-18`) to analyze service utilization trends across developmental stages.
2. **Identifying Service Utilization Trends:**
   - Counts the number of users in each age group, highlighting which groups have higher or lower mental health service utilization.
   - Example Insight: If the `6-12` group has the highest service utilization, it may indicate this age group faces the most mental health challenges or has the best access to services.
3. **Visualizing Disparities:**
   - The bar plot offers an intuitive way to compare age groups, making it easy to identify underserved groups.
   - Example Insight: If the `0-5` group has significantly lower utilization, it could signal barriers in accessing services for young children or a lack of early intervention programs.
4. **Actionable Outcomes:**
   - Helps allocate resources effectively:
     - For example, increasing services for the `13-18` group if they have high utilization but poor outcomes.
   - Guides age-specific intervention strategies:
     - Early intervention for younger groups (`0-5` and `6-12`).
     - Focused programs for adolescents (`13-18`).

Conclusion

This code provides a comprehensive framework for analyzing and visualizing the age distribution of service users. By categorizing and visualizing age groups, it identifies utilization trends and disparities, enabling informed decisions to improve mental health services for children and youth.

Code Explanation in Reference to Scenario 4: Race Group Analysis for Time to Step Down (Snippet #5)

Scenario Context:

Scenario 4 involves analyzing how long different racial groups take to step down from mental health services. "Step down" refers to the process where service users transition from more intensive mental health services to less intensive ones. The goal is to examine the

demographic distribution (race groups) of service users to identify disparities or trends in service utilization or progression across different racial groups.

Code Walkthrough

1. Dataset Loading

- **File Path:**
  - The dataset file, **`time_to_step_down_race_group_cleaned.csv`**, is loaded using pandas into a DataFrame named `time_to_step_down_df`.
- **Error Handling:**
  - A `try-except` block is used to ensure that if the dataset cannot be loaded (e.g., due to file not found or incorrect path), an error message is displayed.
- **Purpose:**
  - Ensures the dataset is available for analysis and prevents the program from crashing in case of errors.

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2. Data Inspection

- **Preview the Data:**
  - The code prints the first few rows of the dataset using **`head()`** to understand its structure and verify the presence of relevant columns like `race_group`.
- **Purpose:**
  - Helps confirm that the dataset contains the necessary data for analysis, including a column specifying race groups (`race_group`) and any other related information.

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3. Visualizing Demographic Distribution

The function **`plot_demographic_distribution()`** is used to visualize the distribution of service users across different race groups.

Step-by-Step Explanation of the Function

1. **Input Parameters:**
   - **`df`:** The dataset (DataFrame) to analyze.
   - **`group_by_column`:** The column in the dataset to group users by (in this case, `race_group`).
   - **`title`:** The title for the visualization.
2. **Column Existence Check:**

    - The function first checks if the specified column (`group_by_column`) exists in the dataset. If it doesn't, an error message is printed, and the function exits.
3. **Visualization with Seaborn:**
    - **`sns.countplot`:** This creates a bar plot of the counts of unique values in the specified column (`race_group`), showing the number of service users in each racial category.
    - **Figure Settings:**
        - A figure size of `(10, 6)` is defined for a clear visualization.
        - The color palette is set to **`viridis`** for a visually appealing gradient effect.
4. **Title and Labels:**
    - The plot is given an appropriate title (`Race Group Distribution for Time to Step Down`).
    - The X-axis is labeled with the race group categories, and the Y-axis represents the count of service users.
5. **Display:**
    - The plot is displayed using **`plt.show()`**.

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4. Plot Output

- The plot will display the **number of service users** for each race group.
- Example Insight:
    - If the `race_group` column contains categories like `White`, `Black`, `Hispanic`, and `Asian`, the plot will show a bar for each category with the count of users in that group.

Relevance to Scenario 4

This code aligns with Scenario 4's objectives by providing a clear demographic analysis of service users based on their race group. Here's how the analysis contributes to the scenario:

1. **Demographic Trends:**
    - By visualizing the number of service users in each race group, the code identifies which groups are more or less represented in the dataset.
    - Example Insight: If one racial group has significantly fewer users, it may indicate barriers to accessing mental health services for that group.
2. **Potential Disparities:**
    - This analysis highlights disparities in service utilization across racial groups.
    - Example Insight: A racial group with very few users might require outreach programs or tailored services to address cultural or systemic barriers.
3. **Step-Down Time Implications:**
    - Although the plot focuses on counts, this is the first step in understanding how race groups are distributed in the dataset. Subsequent analysis could

investigate whether the **time to step down** varies significantly across racial groups.
    - Example Follow-Up: The average or median step-down time for each racial group could be calculated to uncover disparities in service progression.
4. **Actionable Insights:**
    - The visualization helps policymakers and service providers:
        - Identify underserved racial groups.
        - Develop strategies to ensure equitable access and progression through mental health services.

Conclusion

This code snippet provides a foundational demographic analysis for Scenario 4. By visualizing the distribution of service users by race group, it sets the stage for deeper analyses into potential disparities in mental health service outcomes, such as time to step down. The insights gained can drive efforts to address inequities and improve service delivery for all racial groups.

Code Explanation in Reference to Scenario 5: Temporal Trends Analysis (Snippet #6)

Scenario Context:

In Scenario 5, the focus is on analyzing **temporal trends** in the dataset. Temporal trends highlight how the number of service users, as well as their demographic characteristics (age group, race group, and gender), have evolved over time. This type of analysis can reveal patterns or shifts in service utilization, helping decision-makers adjust strategies to meet changing needs.

Code Walkthrough

1. Dataset Loading

- The dataset **`demographics_age_group_cleaned.csv`** is loaded into a DataFrame using pandas.
- The dataset is assumed to include **temporal columns**, such as `'year'` or `'month'`, alongside demographic columns like `'age_group'`, `'race_group'`, and `'sex'`.

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2. Temporal Data Inspection

- **`print(df.head())`:**
  - Prints the first few rows of the dataset to confirm the presence of temporal columns, such as `'year'`, and demographic columns like `'age_group'` and `'race_group'`.

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3. Analyzing Temporal Trends

3.1 Yearly Service Utilization Trends

- **Step 1: Grouping Users by Year**
  - The dataset is grouped by the `'year'` column using **`groupby('year')`**, and the number of users is counted for each year using **`size()`**.
  - A new DataFrame, **`yearly_data`**, is created with columns `'year'` and `'user_count'`.
- **Step 2: Visualizing Service Users Over the Years**
  - A **line plot** is created using **`sns.lineplot`** to show the trend of service utilization over the years.
  - **Purpose:** This plot provides a clear picture of whether service utilization has increased, decreased, or fluctuated over time.
  - **Potential Insights:**
    - A rising trend could indicate increasing awareness and access to services.
    - A declining trend might suggest reduced availability or outreach of services.

3.2 Shifts in Age Group Distribution Over Time

- The dataset is grouped by **`['year', 'age_group']`**, and the number of users in each age group is counted.
- The **`unstack()`** function is used to create a table where each age group becomes a separate column, with years as rows and counts as values.
- A **multi-line plot** is created to visualize the trends for each age group.
- **Purpose:** Shows how different age groups (e.g., children, adolescents) are represented in the dataset over time.
- **Potential Insights:**
  - One age group's increasing trend might indicate growing needs for that demographic.
  - A decreasing trend for another group might highlight areas where services are no longer meeting the demand.

3.3 Shifts in Race Group Distribution Over Time

- The dataset is grouped by `['year', 'race_group']` to count users for each race group per year.
- The `unstack()` function is used to prepare the data for plotting.
- A **multi-line plot** is created to visualize the trends for each race group.
- **Purpose:** Examines how service utilization varies across racial groups over time.
- **Potential Insights:**
  - Growth in service users from one racial group might indicate successful outreach or improved access.
  - Decline for another racial group could highlight barriers to access or underrepresentation.

3.4 Shifts in Gender Distribution Over Time

- The dataset is grouped by `['year', 'sex']` to count users for each gender per year.
- The `unstack()` function is used to format the data for plotting.
- A **multi-line plot** is created to visualize gender trends.
- **Purpose:** Tracks how service utilization varies between genders over time.
- **Potential Insights:**
  - An increasing trend for one gender might indicate growing recognition of mental health needs for that group.
  - A stagnant or declining trend could highlight underserved groups.

Key Outputs and Their Relevance to Scenario 5

1. **Service Users Over Time (Yearly Trends):**
   - **Relevance:** Shows whether overall service demand is increasing or decreasing over time. A sharp increase might indicate rising awareness, while a drop could signal accessibility challenges.
2. **Age Group Trends Over Time:**
   - **Relevance:** Highlights which age groups are utilizing services most. For instance:
     - Increasing demand from teenagers may require expanding adolescent-focused services.
     - A drop in young children's service usage might suggest barriers for that age group.
3. **Race Group Trends Over Time:**
   - **Relevance:** Identifies racial disparities in service utilization. For example:
     - Growth in one racial group might indicate effective outreach.
     - Decline in another group could point to systemic issues or cultural barriers.
4. **Gender Trends Over Time:**

- **Relevance:** Tracks whether both genders are benefiting equally from services. For instance:
  - A consistent rise in male users could reflect breaking down stigma.
  - Stagnation in female users might suggest a lack of targeted services.

Insights Derived from Temporal Analysis in Scenario 5

The temporal trends analysis helps answer the following key questions:

1. **How has overall service utilization changed over time?**
   - Rising or falling trends can guide resource allocation and service planning.
2. **Are certain demographics (age, race, gender) underrepresented or underserved?**
   - Identifying underserved groups ensures equitable access to services.
3. **Which demographics show increasing demand, and how can services adapt?**
   - For example, an increasing trend in adolescents might require more youth-focused programs.

Conclusion

This code provides a robust framework for analyzing temporal trends in service utilization. By visualizing yearly changes and demographic shifts, it allows service providers to understand evolving patterns and tailor mental health services to meet the dynamic needs of different population groups.

Code Explanation in Reference to Scenario 6: Cross-Tabulation of Age Group and Race Group (Snippet #7)

Scenario Context:

In Scenario 6, the focus is on **cross-tabulating age groups and race groups** to explore their relationship and understand how demographics are distributed across different categories. This analysis helps identify patterns or disparities in service utilization among various age groups and racial demographics.

Code Walkthrough

1. Loading Datasets

- Two datasets are involved:
  - `children-and-youth-demographics-by-age-group.csv` (containing data on age group distributions).

    - **`children-and-youth-demographics-by-race-group.csv`** (containing data on race group distributions).
- A dictionary, **`file_mapping`**, maps the file names to more descriptive dataset names (**`demographics_age_group`** and **`demographics_race_group`**) for better readability.
- The code iterates over files in the working directory and loads the ones matching the `file_mapping` keys into the **`datasets`** dictionary using `pandas.read_csv()`.

Key Actions:

- Ensures both datasets are loaded and named appropriately.
- Prints the first few rows of each dataset to confirm successful loading and check the structure.

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2. Merging Datasets

- After loading, the two datasets are merged into a single DataFrame, **`merged_df`**, using `pd.concat()` with **`axis=1`** (side-by-side).
    - **Purpose:** Combines age group and race group demographic data to enable cross-tabulation.

Key Actions:

- Prints the shape and first few rows of the merged dataset to verify the merge.

Possible Adjustments:

- If the datasets have a common key (e.g., `user_id`), **`pd.merge()`** could be used instead of **`pd.concat()`**.

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3. Performing Cross-Tabulation

- Cross-tabulation is performed using **`pd.crosstab()`**, with:
    - **`index='age_group'`**: Rows represent different age groups.
    - **`columns='race_group'`**: Columns represent different race groups.
    - **`margins=True`**: Adds row and column totals for easier interpretation.
- The resulting table shows the frequency of service users for each **age group** and **race group combination**, along with overall totals.

Key Actions:

- Prints the cross-tabulation result to inspect the data.

Relevance to Scenario 6:

- **Cross-tabulation** provides insights into the relationship between age and race demographics:
  - Which race groups are more prominent in specific age groups?
  - Are there any underrepresented or overrepresented demographics?

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4. Visualizing Cross-Tabulation (Heatmap)

- If the cross-tabulation result is not empty:
  - A **heatmap** is created using `sns.heatmap()` to visualize the distribution of age groups vs. race groups.
  - **Key Features:**
    - **Annotations (`annot=True`)**: Displays raw counts in each cell.
    - **Color Mapping (`cmap="YlGnBu"`)**: Shows intensity (higher counts = darker shades).
    - **Excludes Totals (`iloc[:-1, :-1]`)**: Prevents row/column totals from being plotted.
  - **Purpose:** Makes it easier to identify patterns or disparities at a glance.

Key Actions:

- Sets axis labels and rotates the x-axis labels for better readability.

Relevance to Scenario 6:

- The heatmap highlights where certain age groups are overrepresented or underrepresented across race groups:
  - Darker shades = higher representation.
  - Lighter shades = lower representation or underutilization.
- Helps identify trends or disparities visually for targeted interventions.

Key Outputs and Their Relevance to Scenario 6

1. Cross-Tabulation Table

- **Relevance:** Summarizes the distribution of service users across **age groups** and **race groups**.
  - Example Insight: If one race group dominates all age groups, it may indicate overrepresentation or bias in service delivery.
  - If a specific age group (e.g., 0–5 years) is underrepresented across all race groups, it may highlight barriers to access for younger children.

2. Heatmap

- **Relevance:** Visual representation of the cross-tabulation data.
  - Example Insight: Dark regions may show where resources are most utilized (e.g., adolescents from a specific racial background).
  - Light regions might indicate gaps in service utilization, suggesting where targeted outreach is needed.

Insights Derived from Cross-Tabulation in Scenario 6

- **Demographic Disparities:** Identifies if certain race groups dominate specific age groups, indicating potential inequities.
- **Service Gaps:** Highlights underrepresented groups (e.g., fewer users from a race group or age group).
- **Resource Allocation:** Helps policymakers focus resources on underserved age or race groups to ensure equitable service delivery.

Conclusion

This code is tailored for Scenario 6, where understanding the relationship between **age groups** and **race groups** is essential for identifying disparities and informing decision-making. The cross-tabulation and heatmap provide both numerical and visual insights, enabling stakeholders to address equity and representation issues effectively.

Code Explanation in Reference to Scenario 7: Exploring Socioeconomic Barriers

Scenario Context:

In Scenario 7, the focus is on **analyzing socio economic barriers** that may affect service utilization. The goal is to examine the relationships between demographic variables (e.g., age, race, and written language), socioeconomic factors (e.g., income level), and service utilization (e.g., the amount of services accessed). By leveraging this analysis, insights can be drawn to identify disparities and address potential barriers.

Code Walkthrough

1. Loading and Organizing Datasets

- The code first attempts to load three datasets:
  - `demographics_age_group`: Contains demographic information, such as age, race, sex, or ethnicity.

    - `utilization_age_group`: Includes information on mental health service utilization by age group.
    - **Socioeconomic Data** (optional): May include details on income levels, written languages, or other socioeconomic factors.
- These datasets are stored in a dictionary called `datasets`.

Key Actions:

- If a dataset (e.g., socioeconomic data) is unavailable, the code skips its analysis but proceeds with available data.
- Prints column names for each dataset to help identify shared keys for merging.

Relevance to Scenario 7:

- **Demographics, utilization, and socioeconomic data** are essential for understanding barriers and disparities in service access. Missing data may limit the depth of the analysis.

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2. Merging Datasets

- The datasets are merged based on common columns, such as `population_name`, `age_group`, `race_group`, or `income_level`.
- The merged dataset (`merged_df`) combines demographic, utilization, and socioeconomic information.

Key Actions:

- Identifies overlapping columns for merging.
- If socioeconomic data is present, it is integrated with the demographics and utilization data.

Relevance to Scenario 7:

- **Merging datasets** allows for a holistic view of how socioeconomic factors influence service utilization, enabling the identification of barriers like low income or language limitations.

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3. Data Cleaning and Validation

- The merged dataset is checked for missing values.
- Missing values are handled using two strategies:
    - Dropping rows with missing values.
    - Imputing missing values using the median for columns like `TOTAL_CT`, `MHS1_CT`, and `AMOUNT` (service utilization metrics).

Relevance to Scenario 7:

- Ensures the dataset is clean and reliable for analysis.
- Filling missing values prevents biases introduced by dropping rows entirely, especially when dealing with socioeconomic barriers where missing data is common.

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4. Correlation Analysis

- A **correlation matrix** is created for numeric columns (e.g., income levels, service utilization, total counts).
- **Heatmap Visualization**: Displays correlations between variables.

Relevance to Scenario 7:

- Identifies relationships between socioeconomic variables (e.g., income) and service utilization metrics.
- Example Insight: A strong negative correlation between income level and service utilization might indicate that lower-income individuals face access barriers.

---

5. Service Utilization by Income Level

- A **box plot** is generated to compare service utilization (e.g., `AMOUNT`) across income levels.
    - **X-axis:** Income levels (e.g., low, medium, high).
    - **Y-axis:** Service utilization (e.g., amount of services accessed).

Relevance to Scenario 7:

- Visualizes disparities in service utilization across income groups.
- Example Insight: If higher-income groups utilize services significantly more than lower-income groups, it may indicate financial or systemic barriers for low-income individuals.

---

6. Demographics Analysis

- A **cross-tabulation** is performed between **race group** and **income level**, creating a table that shows the count of individuals in each race group for each income level.
- **Heatmap Visualization**: Highlights income disparities across racial groups.

Relevance to Scenario 7:

- Identifies income-based barriers across racial groups.
- Example Insight: If a specific race group has a higher proportion in lower-income levels, it may indicate systemic inequities.

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7. Hypothesis Testing

Two statistical tests are used to quantify the relationships between variables:

(a) Chi-Square Test

- Tests for a significant association between **race group** and **income level** (categorical variables).
- Outputs:
  - **Chi-square statistic**: Measures the strength of association.
  - **p-value**: Determines statistical significance (e.g., $p < 0.05$ indicates a significant relationship).

Relevance to Scenario 7:

- Example Insight: A significant association might indicate that certain race groups are disproportionately represented in specific income levels, suggesting systemic socioeconomic barriers.

(b) ANOVA Test

- Tests for differences in service utilization (e.g., `AMOUNT`) across **income levels** (numerical vs categorical variables).
- Outputs:
  - **F-statistic**: Measures variance across groups.
  - **p-value**: Determines significance.

Relevance to Scenario 7:

- Example Insight: A significant result might indicate that income levels significantly affect service utilization, with lower-income groups potentially accessing fewer services.

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8. Socioeconomic Barriers: Written Language

- Analyzes the impact of **written language** on service utilization:
  - **Bar Plot Visualization**: Compares average service utilization (e.g., `AMOUNT`) across different written languages.

Relevance to Scenario 7:

- Highlights potential language barriers to accessing services.
- Example Insight: If individuals who speak certain languages (e.g., non-English) utilize services less, it may indicate a need for language-specific outreach or support.

Key Outputs and Insights Relevant to Scenario 7

1. Correlation Analysis

- **Output:** Correlation heatmap.
- **Insight:** Identifies socioeconomic variables most strongly correlated with service utilization.

2. Service Utilization by Income

- **Output:** Box plot.
- **Insight:** Highlights disparities in service access across income levels (e.g., higher-income groups accessing more services).

3. Race and Income Cross-Tabulation

- **Output:** Heatmap of race group vs. income level.
- **Insight:** Identifies racial groups disproportionately represented in lower income brackets.

4. Hypothesis Testing

- **Output:** Chi-square and ANOVA test results.
- **Insight:** Quantifies relationships between race, income, and service utilization.

5. Language Barriers

- **Output:** Bar plot of service utilization by written language.
- **Insight:** Highlights underutilization of services by non-native speakers, indicating language as a socioeconomic barrier.

Conclusion

The code effectively addresses Scenario 7 by merging datasets, cleaning data, and analyzing the relationships between socioeconomic factors and service utilization. It highlights barriers such as income, race, and language that may limit access to services. The statistical tests and visualizations provide actionable insights for addressing these disparities and improving equity in service delivery.